

PAPER_204: UQFF Dark Matter — NFW, SIDM, Rotation Curves, and Virial Theorem

Version: 1.0

Date: March 13, 2026

Session: 50 — grok_share_7514fe.txt Full Audit

Author: Star-Magic UQFF Research Framework

Source: grok_share_7514fe.txt lines 6096-6110 (BB_C_Equations items 1326-1340)

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$U_{b_i}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad \beta_i = 0.61$$

Abstract

This paper applies the UQFF buoyancy framework to dark matter structural physics: the Navarro-Frenk-White (NFW) density profile, NFW rotation curve, self-interacting dark matter (SIDM) core formation, virial theorem mass estimation, strong gravitational lensing Einstein radius, void density evolution, and peculiar velocity. The UQFF perspective unifies these through F_UBii operators that embed DM physical expressions into F_rel/E_LEP scaled buoyancy forces, capturing the feedback between vacuum energy and CDM structure formation.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. NFW Density Profile

$$\rho_{\text{NFW}}(r) = \rho_s / ((r/r_s) \cdot (1 + r/r_s)^2)$$

Parameters:

ρ_s = characteristic density (from halo mass-concentration relation)

r_s = scale radius (from NFW fit to N-body simulations)

$c = r_{\text{vir}}/r_s$ = concentration parameter ($c \sim 10$ at $z=0$, increases at higher z)

Mass enclosed:

$$M(r) = 4\pi \rho_s r_s^3 [\ln(1+r/r_s) - r/r_s/(1+r/r_s)]$$

$$F_{\text{UBii,nfw}} = -F_{\text{rel}} \times (\rho_{\text{NFW}}(r) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (4\pi r^2 \cdot d\rho/dr) \times r$$

$$U_{m,\text{nfw}}(r) = \mu(\rho_{\text{vac}}) \cdot (1 - e^{-\rho t}) \cdot [\text{Fit to universal NFW form } \rho_s/(x \cdot (1+x)^2)]$$

Physical context:

NFW is universal for CDM halos (Milky Way to galaxy clusters)

UQFF: vacuum field DPM_grav term deepens the NFW potential well

Core-cusp tension: NFW predicts cusp $\rho \propto r^{-1}$, observations often show cores

2. NFW Rotation Curve

$$v(r)^2 = 4\pi G \rho_s r_s^2 [\ln(1+x) - x/(1+x)] / r \quad x = r/r_s$$

Asymptotic limits:

$$\begin{aligned} r \ll r_s: v(r) &\propto r^{0.5} && \text{(inner rising)} \\ r \sim r_s: v(r) &\sim \text{maximum} && \text{(peak rotation speed)} \\ r \gg r_s: v(r) &\propto r^{-0.5} \cdot \ln(r)^{0.5} && \text{(slowly declining)} \end{aligned}$$

Flat rotation curves require NFW halo + baryons together:

$$v_{\text{total}}^2(r) = v_{\text{bary}}^2(r) + v_{\text{NFW}}^2(r)$$

$$F_{\text{UBii,nfwrot}} = F_{\text{rel}} \times (v^2(r)/G / E_{\text{LEP}}) \times Q_{\text{wave}} \times [\ln(1+x) - x/(1+x)]$$

$$U_{\text{m,nfwrot}}(x) = \mu(\rho_{\text{vac}}) \cdot (1 - e^{-\tau t}) \cdot [\text{Flat rotation for } r \gg r_s]$$

Calibration: Milky Way NFW:

$$\rho_s \sim 0.3 \text{ GeV/cm}^3, r_s \sim 20 \text{ kpc}, v_c \sim 220 \text{ km/s at Solar circle (8 kpc)}$$

3. SIDM Core Formation

Self-interacting DM rate:

$$G = \sigma \cdot (s/m) \cdot v_{\text{rel}} \quad (\text{interaction rate})$$

Core formation timescale:

$$t_{\text{core}} \sim (\sigma \cdot s/m)^{-1} \sim 10^{10} \cdot (\sigma / 108 \text{ Mpc}^3)^{-1} \cdot (s/m / 1 \text{ cm}^2/\text{g})^{-1} \text{ yr}$$

Exponential density evolution:

$$\rho_{\text{core}}(t) = \rho_{\text{init}} \cdot e^{-Gt} \quad (\text{NFW cusp converts to core when } Gt \sim 1)$$

$$F_{\text{UBii,sidm}} = -F_{\text{rel}} \times (G \cdot \rho_{\text{init}} / E_{\text{LEP}}) \times Q_{\text{wave}} \times \ln(0.02N)$$

$$U_{\text{m,sidm}}(t) = \mu(\rho_{\text{vac}}) \cdot (1 - e^{-\tau t}) \cdot [\text{Exponential density flattening}]$$

Observational constraint: $s/m \sim 0.1\text{--}1 \text{ cm}^2/\text{g}$ (galaxy clusters, Bullet Cluster)

Planck does not exclude SIDM at this level (CDM and SIDM nearly identical on large scales)

SIDM predictions:

- Dwarf galaxies: soliton cores of radius $\sim 100 \text{ pc}$ (observed)
 - Galaxy clusters: rounder, less concentrated halos
 - Bullet Cluster: upper limit $s/m < 1.25 \text{ cm}^2/\text{g}$ (from offset DM centroid)
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4. Virial Theorem Mass

$$2K + W = 0 \quad (\text{virial equilibrium for collisionless system})$$

$$K = (3/2)M \cdot s_v^2 \quad (\text{kinetic energy})$$

$$W = -(3/5)GM^2/r_h \quad (\text{potential for uniform sphere})$$

Virial mass:

$$M_{\text{vir}} = 2|K|/G = 3 \cdot s_v^2 \cdot r_h / G \quad (\text{for spherical system})$$

Cluster mass from spectroscopic s_v :

$$M(< r) = 3s_v^2(r) \cdot r/G + \text{corrections for anisotropy + pressure}$$

$$F_{UBii,vir} = F_{rel} \times (M_{vir} / E_{LEP}) \times Q_{wave} \times (s_v^2/G) \times 3$$

$$U_{m,vir}(r) = \mu(\rho_{vac}) \cdot (1 - e^{-\rho t}) \cdot [s_v^2 = GM/(3r)]$$

X-ray virial:

$$M_{vir,X} = 3s_X^2 \cdot r_h/G \quad (\text{from X-ray spectroscopy instead of optical})$$

$$U_{m,virx}(r) = \mu(\rho_{vac}) \cdot (1 - e^{-\rho t}) \cdot [\text{Matches Chandra cluster observations}]$$

Numerical calibration: Coma Cluster

$$s_v \sim 880 \text{ km/s}, r_h \sim 1 \text{ Mpc} \Rightarrow M_{vir} \sim 2 \times 10^{15} M_\odot$$

5. Strong Gravitational Lensing

Einstein radius:

$$\theta_E = \sqrt{4GM(<\theta)/c^2 \cdot D_{LS}/(D_L \cdot D_S)}$$

Critical surface density:

$$S_{cr} = c^2 D_S / (4\pi G D_L \cdot D_{LS})$$

Convergence: $\kappa = S/S_{cr}$

Shear: γ (traceless tidal field)

Multiple images: $\kappa = 1$ at image positions

$$F_{UBii,lens} = F_{rel} \times (\kappa_E / E_{LEP}) \times Q_{wave} \times (S_{cr} \cdot \kappa)$$

$$U_{m,lens}(\theta) = \mu(\rho_{vac}) \cdot (1 - e^{-\rho t}) \cdot [\kappa_E = \kappa(a, \theta) \text{ from lensing equation}]$$

Einstein ring systems:

$$\text{SDP.81 (ALMA): } z_L=0.3, z_S=3.04 \Rightarrow \theta_E \sim 1.5'' \Rightarrow M(<\theta_E) \sim 10^{11} M_\odot$$

$$\text{UQFF: vacuum } \rho \text{ correction to } D_{LS} \text{ shifts } \theta_E \text{ by } \sim 0.1\%$$

6. Void Density Evolution

Void density contrast (spherical top-hat model):

$$d_v(a) = -(3/5) \cdot (0_m \cdot a + 0_\rho)^{-3/2} \cdot d_{v0}$$

d_{v0} = initial void underdensity

Linear theory: $d_v \propto -a^{1/2}$ in ρ -dominated epoch (voids deepen faster)

Shell-crossing: $d_v \propto -1$ marks void edge (no further underdensity growth)

$$F_{UBii,voidden} = -F_{rel} \times (|d_v(a)| / E_{LEP}) \times Q_{wave} \times (0_m \cdot a + 0_\rho)^{-3/2}$$

$$U_{m,voidden}(a) = \mu(\rho_{vac}) \cdot (1 - e^{-\rho t}) \cdot [d \propto a^{-1} \text{ in matter domination}]$$

Physical context in UQFF:

VOIDS are dominated by vacuum energy (ρ) $\Rightarrow$ UQFF's $\rho c^2/3$ term strongest here

F_UBii,voidden predicts void expansion driven by vacuum buoyancy

7. Peculiar Velocity Field

Peculiar velocity from linear theory:

$$v_{\text{pec}}(r) = -(fH/3) \cdot d(r') \cdot r \cdot dr'/r^2 \quad (\text{spherical approximation})$$

Redshift space distortions (RSD):

$$v_{\text{pec,observed}} = f \cdot H \cdot r + \text{noise} \quad (\text{adds to Hubble flow})$$

$$f \sim \Omega_m^{\{0.55\}} \quad (\text{growth rate approximation})$$

Cosmic flow from Laniakea to CMB dipole:

$$v \sim 630 \text{ km/s toward Perseus-Pisces}$$

$$F_{\text{UBii,pec}} = F_{\text{rel}} \times (fH \cdot d(r)/3 / E_{\text{LEP}}) \times Q_{\text{wave}} \times (dv/dz \text{ systematic})$$

$$U_{\text{m,pec}}(r) = \mu(\Omega_{\text{vac}}) \cdot (1 - e^{-\Omega_{\text{vac}} t}) \cdot [\text{Spherical void: integrate Poisson}]$$

8. Cluster Shock Mach Number and Merger Timescale

Shock Mach number from X-ray temperature jump:

$$M = [(\Omega+1)(\Omega^2/\Omega_1) + (\Omega-1)] / (2\Omega) \quad (\text{from Rankine-Hugoniot})$$

$$T_2/T_1 = [2\Omega M^2 - (\Omega-1)] \cdot [(\Omega-1)M^2 + 2] / [(\Omega+1)M]^2 \quad (\text{temperature jump})$$

$$\text{Coma radio relic: } M \sim 2.5 \quad (\text{from spectral index } a = (M^2+1)/(M^2-1))$$

Merger crossing/dynamical timescale:

$$t_{\text{merge}} = r_{\text{vir}}/s_v = v(3r_{\text{vir}}^3/(5GM))$$

$$F_{\text{UBii,mach}} = F_{\text{rel}} \times (M \cdot v_s / E_{\text{LEP}}) \times Q_{\text{wave}} \times (T_2/T_1)$$

$$F_{\text{UBii,merg}} = F_{\text{rel}} \times (t_{\text{merge}} / E_{\text{LEP}}) \times Q_{\text{wave}} \times (r_{\text{vir}}/v_c)$$

$$U_{\text{m,mach}}(\Omega) = \mu(\Omega_{\text{vac}}) \cdot (1 - e^{-\Omega_{\text{vac}} t}) \cdot [\text{Matches Coma radio relic shocks } M \sim 2-3]$$

$$U_{\text{m,merg}}(t) = \mu(\Omega_{\text{vac}}) \cdot (1 - e^{-\Omega_{\text{vac}} t}) \cdot [3r_{\text{vir}}/(5GM)]$$

9. Summary: UQFF DM Force Hierarchy

Scale	Dominant DM Process	F_UBii Variant	Observable
kpc (dwarf)	SIDM core formation	F_UBii,sidm	Kpc-scale DM core
kpc (Milky Way)	NFW rotation	F_UBii,nfwrot	$v_c = 220 \text{ km/s}$
Mpc (clusters)	Virial mass	F_UBii,vir	$s_v = 880 \text{ km/s}$ (Coma)
Mpc (clusters)	Lensing	F_UBii,lens	$\Omega_E \sim 1'$ for clusters

Scale	Dominant DM Process	F_UBii Variant	Observable
100 Mpc (voids)	Void underdensity	F_UBii,voidden	d ~ -0.8 in big voids
Bulk (cos- mological)	Peculiar velocity	F_UBii,pec	630 km/s Laniakea flow

10. References

- grok_share_7514fe.txt lines 6096-6110 (BB_C_Equations items 1326-1340, 1262-1268)
- PAPER_199: F_UBii Taxonomy Part 2 (cosmological)
- PAPER_200: Um Universal Magnetism Catalogue
- Navarro, Frenk, White 1996, 1997
- Spergel & Steinhardt 2000 (SIDM)
- Planck 2015 lensing